LNS alone does not imply that the utility constraint binds in (EM)

Let $L = 2$, $X = \mathbb{R}^2_+$ and let preferences be represented by

$$u(x) = \begin{cases} x_1 + x_2 & \text{if} \quad x_1 + x_2 < 1 \\ x_1 + x_2 + 1 & \text{if} \quad x_1 + x_2 > 1 \\ 1 + x_2 & \text{if} \quad x_1 + x_2 = 1. \end{cases}$$

The indifference sets look the same as for perfect substitutes except that the line $x_1 + x_2 = 1$ is not an indifference set: that line resembles lexicographic preferences in that, the more x_2, the better along this line. These preferences are certainly LNS—but not continuous. The figure illustrates indifference curves and the region $x_1 + x_2 = 1$.

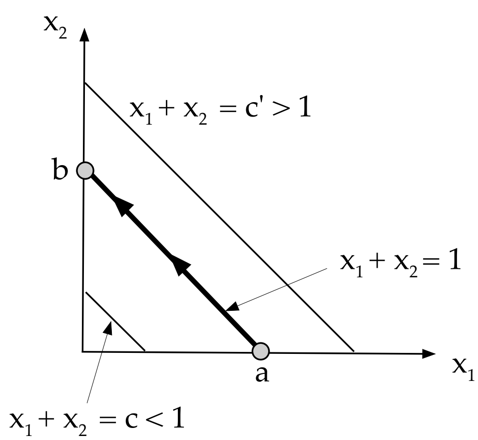

Figure 1: Indifference sets for the example. The lines $x_1 + x_2 = c$ for $c > 1$ and $c < 1$ are indifference curves. Along the line $x_1 + x_2 = 1$, bundles with more good 2 are better, as indicated by arrows along the bold line. Set $\bar{u} = 1$ and $p = (1, 1)$. The point $a = (1, 0)$ is the unique bundle with $u(x) = 1$. The point $a \in h(p, \bar{u})$; but also $b = (0, 1) \in h(p, 1)$ with $u(b) = 2 > 1 = \bar{u}$. The point b minimizes expenditure, but the utility constraint is slack.

Now consider $p_1 = p_2 = 1$ and $\bar{u} = 1$. The unique point with utility 1 is $(1, 0)$ and that point solves the EM problem and costs 1. But $(0, 1)$ also costs 1 with $u(0, 1) = 2 > 1 = \bar{u}$: these representable, LNS, but discontinuous preferences can have expenditure minimizing points for which the utility constraint is slack. In short, continuity alone implies the utility constraint binds, but LNS alone does not.